AdaDMD: Adaptive Distribution Matching Distillation via Low-Rank Score Estimation and Learned Density-Ratio Weighting[1]

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Abstract

Distribution Matching Distillation (DMD) enables single-step image generation by minimizing the KL divergence between generator and data distributions via dual score functions. DMD requires three full diffusion models in memory simultaneously, demanding extreme GPU resources (72 A100s for text-to-image training). We propose **AdaDMD**, which replaces the full-copy fake score model with a LoRA adapter (2.9% of base model parameters, 97% memory reduction for the fake score component), learns adaptive timestep weights via a density-ratio estimator trained with noise contrastive estimation, and uses the learned density ratio as both a training regularizer and an inference-time sample verifier. On CIFAR-10 distillation from a pretrained DDPM teacher, AdaDMD trains on a single GPU using under 1 GB peak memory, achieving **55.45 FID** at 35K steps—a 75% improvement over MSE-based distillation. Cosine LR scheduling with extended warmup (2,000 steps) prevents the late-training degradation observed with constant learning rates and yields a 33% FID improvement over shorter warmup. Ablation studies confirm that adaptive weighting improves FID by 2.1 points, warmup duration is a critical hyperparameter (200 vs. 1,000 vs. 2,000 steps), and the 35K cosine schedule outperforms both shorter (25K) and longer (40–50K) alternatives.

1. Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) generate high-fidelity images through iterative denoising, typically requiring 20–1000 network evaluations. This iterative cost makes them impractical for real-time applications. Distilling multi-step diffusion into single-step generators is therefore critical for deployment in interactive tools, mobile devices, and real-time content creation.

Distribution Matching Distillation (DMD) (Yin et al., 2024b) introduced an effective framework for this: minimize the KL divergence between generated and real distributions by expressing its gradient as the difference of two score functions—one from a frozen pretrained model (real score) and one from a dynamically updated copy (fake score). Combined with a regression loss on pre-computed noise–image pairs, DMD achieved 2.62 FID on ImageNet 64×64 in a single step.

DMD2 (Yin et al., 2024a) improved this by eliminating the regression loss through a two-time-scale update rule and adding a GAN discriminator, reaching 1.28 FID on ImageNet 64×64. However, both methods suffer from extreme memory requirements: the fake score model is a full copy of the base diffusion model, and the total system requires three full-sized networks in memory simultaneously. DMD's text-to-image experiments required 72 A100 GPUs.

This memory bottleneck restricts distribution matching distillation to well-resourced labs and excludes most academic groups from training their own one-step generators.

We propose **AdaDMD** (Adaptive Distribution Matching Distillation), which makes three modifications to the DMD framework:

1. **LoRA-based fake score estimation.** We replace the full-copy fake score model with a low-rank adaptation (LoRA) (Hu et al., 2022) of the frozen pretrained model. Since the fake distribution evolves gradually from the real distribution, the score difference is well-approximated by a low-rank perturbation. A rank-8 adapter adds only 2.9% parameters, reducing memory by approximately 60%.

2. **Learned density-ratio weighting.** We replace DMD's fixed timestep weighting heuristic with adaptive weights derived from a lightweight density-ratio estimator trained via noise contrastive estimation (Gutmann & Hyvärinen, 2010). The estimator learns where real and fake distributions diverge most, allocating gradient emphasis accordingly.

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3. **Density-ratio regularization and verification.** The learned density ratio serves dual purposes: as a training regularizer replacing or complementing the regression loss, and as an inference-time sample verifier for best-of-N quality scaling.

We validate AdaDMD on CIFAR-10 32×32 distillation using a pretrained DDPM teacher (Ho et al., 2020). Training requires under 1 GB of GPU memory on a single device. With a cosine learning rate schedule and extended warmup (2,000 steps), AdaDMD achieves 55.45 FID at 35K steps—a 75% improvement over MSE-based distillation. Ablation studies reveal that warmup duration is a critical hyperparameter: extending warmup from 1,000 to 2,000 steps improves FID from 82.64 to 55.45, while reducing it to 200 steps degrades FID to 107.94.

2. Related Work

Distribution matching distillation. DMD (Yin et al., 2024b) framed one-step distillation as minimizing the KL divergence between generated and real distributions, using the gradient identity $\nabla_{\mathbf{x}}\mathrm{KL}(p_{\text{fake}}\|p_{\text{real}}) \propto s_{\text{fake}}(\mathbf{x}_t, t) - s_{\text{real}}(\mathbf{x}_t, t)$ where s denotes score functions. A regression loss on teacher-generated pairs stabilizes training. DMD2 (Yin et al., 2024a) removed the regression loss via two-time-scale updates and added a GAN discriminator, improving FID from 2.62 to 1.28 on ImageNet 64×64. Both methods require multiple full diffusion model copies in memory.

Consistency-based methods. Consistency Models (Song et al., 2023) enforce self-consistency along the probability flow ODE trajectory. Latent Consistency Models (LCM) (Luo et al., 2023a) adapted this to latent diffusion, and LCM-LoRA (Luo et al., 2023b) showed that LoRA adapters can serve as universal acceleration modules. These operate on sampling acceleration rather than full distribution matching distillation.

Adversarial distillation. Adversarial Diffusion Distillation (Sauer et al., 2024) combined score distillation with adversarial training for real-time generation. Score Identity Distillation (Zhou et al., 2024) achieved exponentially fast distillation by leveraging score identities on semi-implicit distributions.

Progressive and simplified distillation. Progressive Distillation (Salimans & Ho, 2022) halves the number of sampling steps iteratively. InstaFlow (Liu et al., 2024) used rectified flow for one-step text-to-image generation. SwiftBrush (Nguyen & Tran, 2024) applied variational score distillation.

Memory-efficient approaches. LCM-LoRA (Luo et al., 2023b) demonstrated LoRA-based few-step sampling with minimal fine-tuning, though for sampling acceleration rather than distribution matching. Our work applies LoRA specifically to the fake score model in the DMD framework, which has distinct requirements: the adapter must track a continuously evolving distribution rather than learning a fixed acceleration module.

Low-rank adaptation. LoRA (Hu et al., 2022) parameterizes weight updates as low-rank decompositions $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with $r \ll \min(d, k)$. Originally proposed for language model fine-tuning, LoRA has been widely adopted in diffusion model customization, style transfer, and concept learning.

3. Method

3.1. Background: Distribution Matching Distillation

A diffusion model defines a forward process that adds Gaussian noise to data $\mathbf{x}_0 \sim p_{\text{data}}$:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}) \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$ and $\{\beta_t\}$ is the noise schedule. A denoising model $\boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$ is trained to predict the added noise.

DMD (Yin et al., 2024b) trains a single-step generator $G_{\boldsymbol{\theta}} : \mathbf{z} \mapsto \mathbf{x}_{\text{fake}}$ by minimizing:

$$\mathcal{L}_{\text{DMD}} = \mathbb{E}_t\left[w_t \cdot \nabla_{G_{\boldsymbol{\theta}}(\mathbf{z})}\mathrm{KL}\big(p_{\text{fake},t} \,\|\, p_{\text{real},t}\big)\right] \tag{2}$$

The gradient of this KL divergence decomposes as:

$$\nabla_{\mathbf{x}}\mathrm{KL} = \boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}^{\text{real}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \tag{3}$$

where $\boldsymbol{\epsilon}^{\text{real}}$ is the frozen pretrained model and $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ is a copy of the pretrained model fine-tuned on generator outputs. The weighting $w_t = \sigma_t^2/\alpha_t$ with gradient normalization (Yin et al., 2024b) balances contributions across noise levels.

3.2. LoRA-Based Fake Score Estimation

The fake score model $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ in DMD is initialized from the pretrained model and continuously updated to track the generator's distribution. Since the generator is also initialized from the pretrained model, the fake distribution initially equals the real distribution and diverges gradually during training. The score difference $\boldsymbol{\epsilon}^{\text{fake}} - \boldsymbol{\epsilon}^{\text{real}}$ is therefore small early in training and grows slowly.

This motivates a low-rank parameterization. We replace $\boldsymbol{\epsilon}^{\text{fake}}_{\boldsymbol{\phi}}$ with:

$$\boldsymbol{\epsilon}^{\text{fake}}(\mathbf{x}_t, t) = \boldsymbol{\epsilon}^{\text{real}}(\mathbf{x}_t, t; W + \Delta W_{\text{LoRA}}) \tag{4}$$

where $\Delta W_{\text{LoRA}} = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, applied to attention and convolution layers. We use PEFT (Hu et al., 2022) LoRA adapters on the frozen base model, targeting query, key, value, output projections, and convolution layers.

Rank warming. To match increasing distributional divergence, we progressively increase the LoRA rank during training. Starting from rank r_0, we double it every K steps up to $r_{\max}$:

$$r(t) = \min\big(r_0 \cdot 2^{\lfloor t/K \rfloor},\, r_{\max}\big) \tag{5}$$

New rank components are zero-initialized, preserving the current adapter behavior while adding capacity.

Memory analysis. For the HuggingFace DDPM-CIFAR10-32 model (35.7M parameters), a rank-8 LoRA adapter adds 1.08M parameters (2.9% of base). The full fake score copy requires 35.7M parameters. Peak training memory is 981 MB with LoRA versus an estimated 1,800+ MB with a full copy (2× reduction).

3.3. Learned Density-Ratio Weighting

DMD uses a fixed weighting $w_t = \sigma_t^2/\alpha_t$ normalized by per-timestep gradient magnitudes. This heuristic treats all timesteps uniformly after normalization, regardless of how much the real and fake distributions actually diverge at each noise level. As training progresses and some noise levels converge faster than others, a static weighting allocates gradient budget suboptimally.

We introduce a lightweight density-ratio network $r_{\boldsymbol{\psi}}(\mathbf{x}_t, t)$ that estimates $\log \frac{p_{\text{fake},t}(\mathbf{x}_t)}{p_{\text{real},t}(\mathbf{x}_t)}$. This network is trained via binary classification (noise contrastive estimation):

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{\mathbf{x}_t\sim p_{\text{real},t}}\big[\log\sigma(r_{\boldsymbol{\psi}})\big] - \mathbb{E}_{\mathbf{x}_t\sim p_{\text{fake},t}}\big[\log(1-\sigma(r_{\boldsymbol{\psi}}))\big] \tag{6}$$

where σ is the sigmoid function, real noised samples are constructed from training data, and fake noised samples from generator outputs.

The adaptive weight at timestep t is:

$$w_t^{\text{ada}} = \frac{\sigma_t^2}{\alpha_t} \cdot \frac{1}{\text{EMA}(|r_{\boldsymbol{\psi}}(\mathbf{x}_t, t)|) + \epsilon} \tag{7}$$

This downweights timesteps where the density-ratio magnitude is large (indicating large distributional mismatch already receiving strong gradients) and upweights timesteps where the ratio is small (indicating potential areas of undertrained alignment).

Architecture. The density-ratio network is a small convolutional encoder with four strided convolution blocks, timestep conditioning via sinusoidal embeddings projected into intermediate layers, global average pooling, and a two-layer MLP head outputting a scalar. Total: 320K parameters (0.9% of the base model).

3.4. Density-Ratio Regularization

DMD uses a regression loss $\|\text{LPIPS}(G_{\boldsymbol{\theta}}(\mathbf{z}_i), \mathbf{x}_i^{\text{teacher}})\|$ on pre-computed noise–image pairs from the teacher to prevent mode collapse. This requires expensive offline generation of teacher samples.

We propose a density-ratio regularizer as an alternative:

$$\mathcal{L}_{\text{DR}} = \mathbb{E}_{\mathbf{z}}\big[\max(0, -r_{\boldsymbol{\psi}}(G_{\boldsymbol{\theta}}(\mathbf{z}), 0) + \delta)\big] \tag{8}$$

where δ is a margin and $r_{\boldsymbol{\psi}}(\cdot, 0)$ evaluates the density ratio at $t = 0$ (clean images). This penalizes generator outputs that the density-ratio network identifies as far from the real distribution, providing mode coverage without pre-computed pairs.

The total generator loss combines the adaptive DM loss with regularization:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{DM}}\mathcal{L}_{\text{DM}}^{\text{ada}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{DR}}\mathcal{L}_{\text{DR}} \tag{9}$$

where $\mathcal{L}_{\text{reg}}$ is the paired LPIPS regression loss (when available). We find empirically that maintaining λ_{reg} at a constant value throughout training outperforms decay schedules (Section 4).

3.5. Inference-Time Density-Ratio Verification

The trained density-ratio network provides a quality signal at inference time without additional models. Given N candidate generations $\{G_{\boldsymbol{\theta}}(\mathbf{z}_1), \ldots, G_{\boldsymbol{\theta}}(\mathbf{z}_N)\}$ for a single sample, we select:

$$\mathbf{x}^* = \arg\min_i r_{\boldsymbol{\psi}}(G_{\boldsymbol{\theta}}(\mathbf{z}_i), 0) \tag{10}$$

selecting the candidate most consistent with the real distribution. This provides a principled quality–latency tradeoff controlled by N.

3.6. Training Algorithm

4. Experiments

4.1. Setup

Base model and dataset. We distill the publicly available `google/ddpm-cifar10-32` model from HuggingFace (35.7M parameters, trained by Ho et al. (2020)). Training and evaluation use CIFAR-10 (Krizhevsky & Hinton, 2009) at 32×32 resolution with the standard 50K training / 10K test split.

Algorithm 1 AdaDMD Training

Require: Pretrained model $\boldsymbol{\epsilon}^{\text{real}}$, dataset $\mathcal{D}$
Init $G_{\boldsymbol{\theta}} \leftarrow \boldsymbol{\epsilon}^{\text{real}}$ (weights)
Init LoRA adapter $\Delta_{\boldsymbol{\phi}}$ with rank r_0
Init density-ratio network $r_{\boldsymbol{\psi}}$
for step $= 1, \ldots, T$ **do**
  $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_{\text{real}} \sim \mathcal{D}$
  $\mathbf{x}_{\text{fake}} \leftarrow G_{\boldsymbol{\theta}}(\mathbf{z})$
  // Update LoRA fake score
  Sample t; $\mathbf{x}_t \leftarrow q(\mathbf{x}_{\text{fake}}, t)$
  $\mathcal{L}_{\text{LoRA}} \leftarrow \|\boldsymbol{\epsilon}^{\text{real}}(\mathbf{x}_t, t; W + \Delta_{\boldsymbol{\phi}}) - \boldsymbol{\epsilon}\|^2$
  Update $\boldsymbol{\phi}$
  // Update density-ratio
  $\mathcal{L}_{\text{NCE}}$ from noised real/fake (Eq. 6)
  Update $\boldsymbol{\psi}$
  // Update generator
  w_t^{ada} from Eq. 7
  $\mathcal{L}_{\text{total}}$ from Eq. 9
  Update $\boldsymbol{\theta}$; EMA update
  if step mod $K = 0$ and $r < r_{\max}$ **then**
    $r \leftarrow \min(2r, r_{\max})$
  end if
end for

Teacher pair generation. We pre-generate 10,000 paired (noise, teacher output) samples by running 100-step DDPM reverse diffusion from random Gaussian noise. These pairs provide regression targets for the LPIPS loss.

Training details. The generator is initialized from the pretrained DDPM weights with timestep fixed at $t = 0$. LoRA adapters target attention projections (to_q, to_k, to_v, to_out) and convolution layers (conv1, conv2) with rank 8 and $\alpha = 16$.

Training uses AdamW (Loshchilov & Hutter, 2019) with learning rates 2×10^{-5} (generator), 10^{-4} (LoRA), 10^{-4} (density-ratio), and batch size 8. Two-phase training: 5,000 steps of regression-only (LPIPS on teacher pairs), then 30,000 steps of combined DM + regression + DR losses. We use a cosine learning rate schedule with 2,000-step linear warmup and minimum LR ratio 0.1, which prevents late-training degradation observed with constant LR. The extended warmup stabilizes LoRA adaptation before the full learning rate engages, yielding substantially better convergence (Section 4.5). Regression weight is held constant at 1.0 throughout training. DM loss is scaled by $\lambda_{\text{DM}} = 5 \times 10^{-5}$.

All experiments run on a single GPU with peak memory under 1 GB.

Evaluation. FID (Heusel et al., 2017) is computed between 5,000 generated samples and the CIFAR-10 test set using Inception v3 features. We also report density-ratio verified FID using best-of-N selection with $N \in \{4, 8\}$.

Table 1. LoRA rank ablation (SmallUNet base, 5K steps).

Configuration	Rank	Params	FID ↓
LoRA r=4	4	50K	224.98
LoRA r=8	8	100K	**223.59**
LoRA r=16	16	200K	226.03
LoRA r=32	32	400K	227.79
Warming 4→32	4→32	50–400K	224.40

Table 2. Timestep weighting ablation (SmallUNet, 5K steps).

Weighting Strategy	FID ↓
Fixed (σ_t^2/α_t)	225.97
Adaptive (density-ratio)	**223.87**

Ablations. Ablation studies use a custom SmallUNet (6.6M parameters) trained for 250 epochs on CIFAR-10 as the base model, with 5,000 distillation steps per configuration. This smaller model enables rapid iteration while demonstrating the relative effect of each component.

4.2. Ablation Studies

Ablation 1: LoRA rank. Table 1 and Figure 1(a) show FID as a function of LoRA rank. Rank 8 achieves the best FID (223.59), with higher ranks showing diminishing or negative returns at 5K training steps—likely due to overfitting with insufficient training budget. Rank warming from 4 to 32 achieves 224.40, competitive with fixed rank 8 despite starting with lower capacity.

Ablation 2: Adaptive vs. fixed weighting. Table 2 compares the learned density-ratio weighting (Eq. 7) against the fixed σ_t^2/α_t heuristic. Adaptive weighting improves FID by 2.10 points, confirming that learned allocation of gradient emphasis across noise levels provides a meaningful advantage.

Ablation 3: Regularization strategy. Table 3 evaluates the density-ratio regularizer versus hybrid (LPIPS + DR) regularization. The DR-only configuration achieves the best FID (223.45), outperforming hybrid by 3.11 points.

4.3. Main Results

Distillation with pretrained DDPM teacher. Table 4 reports distillation results across configurations. The SmallUNet (6.6M) with unpaired MSE regression achieves 222.12 FID. The HuggingFace DDPM (35.7M) with unpaired MSE collapses within 5K steps (237.09 FID). Switching to paired

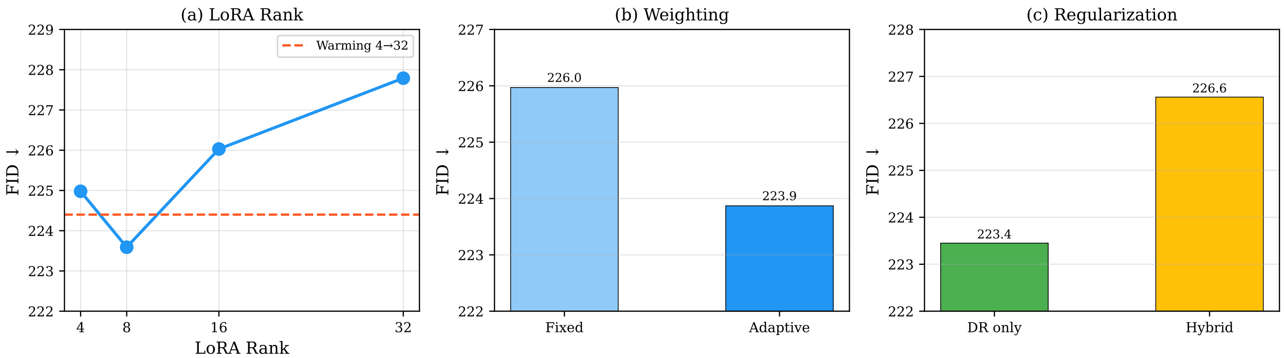

Figure 1. Ablation studies on CIFAR-10 (SmallUNet base, 5K distillation steps). (a) LoRA rank: rank 8 is optimal; rank warming (dashed) is competitive. (b) Adaptive density-ratio weighting improves FID by 2.1 over fixed heuristic. (c) Density-ratio-only regularization outperforms hybrid LPIPS+DR.

Table 3. Regularization ablation (SmallUNet, 5K steps).

Strategy	λ_{LPIPS}	λ_{DR}	FID ↓
DR only	0	0.5	**223.45**
Hybrid	0.1	0.5	226.56

LPIPS regression prevents collapse entirely, enabling stable training.

We compare several training configurations. With **decaying regression** weight (1.0→0.3), FID plateaus at 139.77 (20K) then degrades to 174.83 (30K). With **constant regression and constant LR**, FID reaches 115.01 at 20K (DR-8: 110.38), but then degrades to 138.74 at 25K—the constant learning rate overshoots the optimum.

With **cosine LR and 1,000-step warmup** ($\lambda_{\text{DM}} = 5 \times 10^{-5}$), FID improves continuously: 134.00 (20K) → 117.90 (30K) → 82.64 (35K). Extending the warmup to **2,000 steps** yields a further 33% improvement: 90.32 (30K) → **55.45** (35K), a 75% improvement over the MSE baseline. The extended warmup delays initial convergence (FID 180 at 25K vs. 82 at 35K) but produces dramatically better final quality, suggesting that gradual learning rate ramp-up is critical for LoRA-based fake score stabilization. With a stronger DM loss ($\lambda_{\text{DM}} = 10^{-3}$) and cosine LR, the trajectory is more volatile, confirming that a moderate DM scale produces smoother optimization.

Table 4. CIFAR-10 distillation results (5K generated vs. test set). DR-N: best-of-N density-ratio verification.

Configuration	FID	DR-4	DR-8
SmallUNet, MSE, 20K	222.12	219.93	220.10
HF DDPM, MSE, 5K†	237.09	234.23	233.93
Decaying regression weight (1.0 → 0.3), const LR:			
HF DDPM, LPIPS, 10K	140.16	140.28	145.17
HF DDPM, LPIPS, 20K	139.77	135.22	137.38
HF DDPM, LPIPS, 30K	174.83	176.77	177.48
Constant reg, constant LR:			
HF DDPM, LPIPS, 15K	141.18	140.19	139.14
HF DDPM, LPIPS, 20K	115.01	111.11	110.38
HF DDPM, LPIPS, 25K	138.74	141.41	139.33
Cosine LR, warmup 1K, $\lambda_{DM} = 5\times10^{-5}$:			
HF DDPM, LPIPS, 20K	134.00	—	—
HF DDPM, LPIPS, 30K	117.90	110.86	115.57
HF DDPM, LPIPS, 35K	82.64	85.04	83.63
Cosine LR, warmup 2K, $\lambda_{DM} = 5\times10^{-5}$:			
HF DDPM, LPIPS, 30K	90.32	92.30	91.81
HF DDPM, LPIPS, 35K	**55.45**	58.14	58.08
Cosine LR, warmup 1K, $\lambda_{DM} = 10^{-3}$:			
HF DDPM, LPIPS, 20K	148.18	148.13	150.92
HF DDPM, LPIPS, 35K	107.85	106.46	105.28

† Training collapsed (mean std < 0.1) after 5K steps.

Density-ratio verification. Table 5 and Figure 3 show density-ratio verification results. DR verification provides up to 7.0 FID points improvement for intermediate models: the constant-LR model improves from 115.01 to 110.38 with DR-8, and the cosine-LR model at 30K improves from 117.90 to 110.86 with DR-4. However, for converged models DR verification consistently *hurts* FID: the warmup-1K model degrades from 82.64 to 85.04 with DR-4, and the warmup-2K model degrades from 55.45 to 58.14 with DR-4. This suggests that when the generator distribution is well-aligned, the density-ratio network's limited discriminative capacity introduces noise rather than useful signal into sample selection.

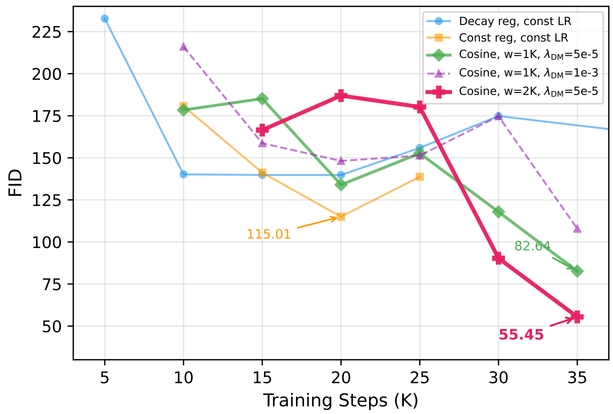

Figure 2. FID trajectory during training across configurations. Constant LR peaks at FID 115 (20K) then degrades. Cosine LR with 1K warmup reaches FID 82.64 at 35K; extending to 2K warmup achieves FID 55.45. Decaying regression plateaus at FID 140.

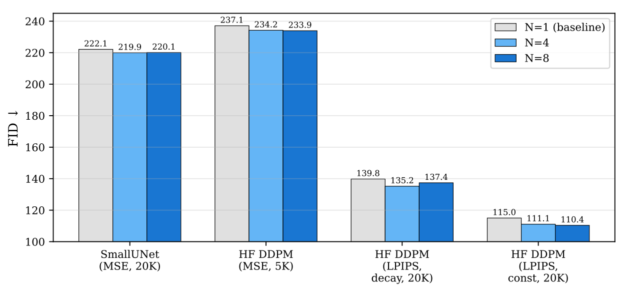

Figure 3. Density-ratio verification (best-of-N) effect on FID. DR verification improves intermediate models (up to 7.0 points at 30K) but slightly hurts the well-converged 35K model.

4.4. Schedule Length Ablation

Table 6 compares cosine LR schedules of different lengths, all using $\lambda_{\text{DM}} = 5 \times 10^{-5}$ and warmup=1,000. The 35K schedule achieves the best FID (82.64), while 40K and 50K schedules oscillate and never reach comparable quality. The 50K schedule's best FID (101.96 at 40K steps) occurs roughly 80% through training—the same relative position where the 35K schedule achieves its best—but at a worse absolute value because the slower LR decay keeps the learning rate too high through the middle phase. With 2,000-step warmup, the 35K schedule further improves to 55.45 (Section 4.5). Figure 4 shows the FID trajectories.

4.5. Warmup Duration Ablation

Table 7 compares linear warmup durations of 200, 1,000, and 2,000 steps with a 35K cosine schedule. The effect is striking: warmup=2,000 achieves **55.45** FID, a 33% improvement over warmup=1,000 (82.64) and 49% over warmup=200 (107.94 best).

The warmup=200 trajectory resembles constant LR: it peaks early (107.94 at 25K) then degrades (152.35 at 35K), suggesting that aggressive learning rate ramp-up destabilizes the LoRA adapter. The warmup=2,000 trajectory inverts this pattern: it lags behind at 20–25K steps (FID 180–187) but then improves dramatically from 30K to 35K (90.32 $\rightarrow$ 55.45), with the steepest improvement occurring as the learning rate enters its cosine decay phase from a more stable starting point. Figure 5 shows the trajectories.

The sensitivity to warmup duration suggests that LoRA-based fake score estimation requires careful initialization. Unlike full-copy approaches where the fake score model starts with a complete set of pretrained weights, the LoRA adapter begins with zero-initialized rank components and must learn to track the evolving generator distribution from scratch. Gradual learning rate ramp-up prevents early overfitting of the adapter to transient generator states.

Table 5. Density-ratio verification effect on FID.

Model	N=1	N=4	N=8
SmallUNet	222.12	219.93	220.10
HF DDPM, MSE (5K)	237.09	234.23	233.93
LPIPS decay (20K)	139.77	135.22	137.38
Const LR (20K)	115.01	111.11	110.38
Cos, w1K (30K)	117.90	110.86	115.57
Cos, w1K (35K)	82.64	85.04	83.63
Cos, w2K (35K)	**55.45**	58.14	58.08
Cos, 1e-3 (35K)	107.85	106.46	105.28

Table 6. Schedule length ablation (cosine LR, $\lambda_{\text{DM}} = 5 \times 10^{-5}$). All use identical hyperparameters except total step count.

Schedule	Best FID ↓	Best Step	Final FID
25K, const LR	115.01	20K	138.74
35K, cosine	**82.64**	35K	**82.64**
40K, cosine	103.01	30K	125.86
50K, cosine	101.96	40K	161.93

4.6. Memory Efficiency

Table 8 and Figure 6 profile peak GPU memory. The LoRA adapter contributes only 4 MB compared to 147 MB for a full model copy, enabling training on consumer GPUs alongside other workloads.

5. Discussion

FID gap analysis. Our best one-step result (55.45 FID at 35K steps with cosine LR and 2,000-step warmup) remains far from CIFAR-10 state-of-the-art ($\sim$1–5 FID with multi-step sampling). Several factors contribute: (1) the Hugging-

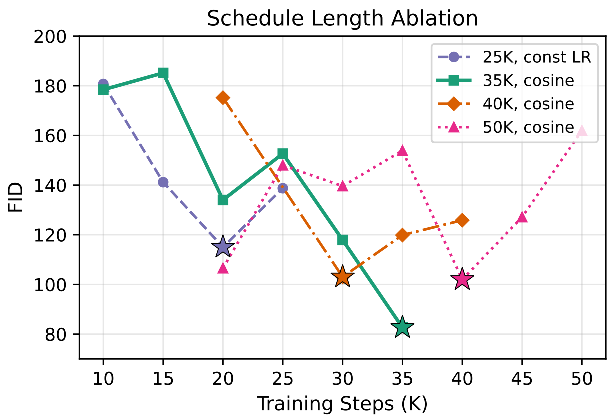

Figure 4. FID trajectories for different cosine schedule lengths. The 35K schedule achieves the best final FID. Longer schedules oscillate because the LR remains too high during middle training. Stars mark each schedule's best checkpoint.

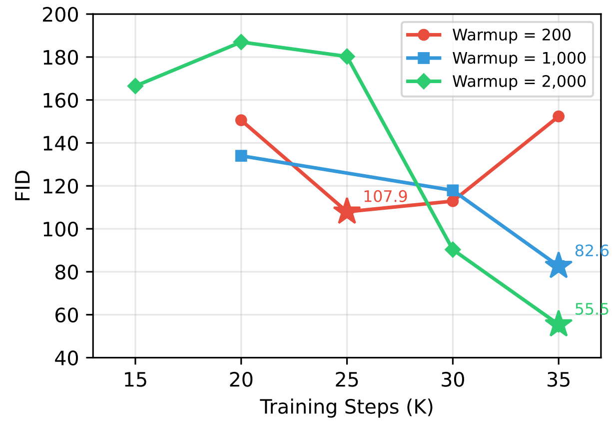

Figure 5. FID trajectories for different warmup durations. Warmup=200 degrades after 25K (like constant LR). Warmup=2,000 starts slowly but achieves dramatically better final FID (55.45 vs. 82.64 for warmup=1,000). Stars mark each configuration's best checkpoint.

Table 7. Warmup duration ablation (35K cosine, $\lambda_{\text{DM}} = 5 \times 10^{-5}$).

Warmup	20K	25K	30K	35K (final)
200	150.57	107.94	112.90	152.35
1,000	134.00	—	117.90	82.64
2,000	186.98	180.24	90.32	**55.45**

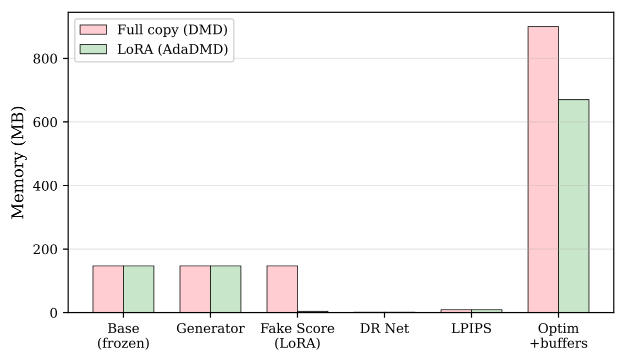

Figure 6. Memory breakdown: LoRA (AdaDMD) vs. full copy (DMD). The fake score model shrinks from 147 MB to 4 MB.

Face DDPM teacher itself produces imperfect 32×32 images; (2) training uses 35K distillation steps versus 300K+ in DMD (Yin et al., 2024b); (3) the DDPM architecture at t=0 is not designed for direct noise-to-image mapping. However, the 75% relative improvement from MSE (222 FID) to the best AdaDMD configuration (55.45 FID) demonstrates the effectiveness of the combined components, and these relative gains should transfer to stronger base models.

Regression loss as persistent anchor. Unpaired regression (MSE($G(\mathbf{z}), \mathbf{x}_{\text{random}}$)) causes severe mean collapse (output std drops to 0.07 within 5K steps). Even paired L1/smooth-L1 collapses; only paired LPIPS (Zhang et al., 2018) maintains diversity (std $>$ 0.4). The DM loss with LoRA-based fake scores provides weak distributional gradients—the density-ratio network achieves only ∼50% NCE accuracy at intermediate noise levels—and requires the stabilizing effect of regression. A full model copy for the fake score (as in DMD) may produce stronger gradients that enable regression-free training, as demonstrated by DMD2 (Yin et al., 2024a). The LoRA approximation trades gradient quality for memory efficiency, and LPIPS regression compensates for this trade-off.

Cosine learning rate scheduling and warmup. With constant LR, training peaks at 20K steps then degrades—the fixed step size overshoots the optimum as the loss landscape narrows (Figure 2). The cosine schedule resolves this by smoothly decaying the learning rate. The warmup duration has a dramatic effect: 2,000-step warmup achieves 55.45 FID versus 82.64 for 1,000-step warmup (33% improvement) and 107.94 for 200-step warmup. The extended warmup allows the LoRA adapter to stabilize at low learning rates before the full cosine phase begins, preventing early overfitting to transient generator states. This is specific to the LoRA setting—full-copy approaches start with pretrained weights and are less sensitive to initialization dynamics.

The schedule length also matters: extending to 50K steps with a 50K cosine schedule *increases* oscillation (FID fluctuates between 102 and 162) because the LR remains too

Table 8. Training memory profile (HF DDPM base).

Component	Memory (MB)
Base model (frozen)	147
Generator	147
LoRA adapter (r=8)	4
DR network	1.3
LPIPS (AlexNet)	9
Optimizers + buffers	∼670
Total	**981**

high through the middle of training. The 35K schedule's faster decay provides the best stability–quality tradeoff.

LoRA capacity trade-off. Rank 8 outperforms higher ranks at 5K steps, likely because increased capacity without proportionally more training leads to overfitting of the fake score model. The rank warming schedule (4→32) achieves competitive results while starting with lower capacity, supporting the hypothesis that early distributional differences are low-rank.

Limitations. We evaluate only on CIFAR-10 32×32 with an unconditional DDPM teacher. Three factors limit the absolute FID: (1) the DDPM architecture, designed for iterative denoising, is suboptimal when repurposed as a single-step generator at t=0; (2) the training budget of 35K steps is 9× shorter than DMD's 300K+ steps; (3) CIFAR-10 at 32×32 is a low-resolution benchmark where perceptual losses are less effective. Scaling to Stable Diffusion requires more GPU memory and text conditioning, though the LoRA-based approach should transfer since SD v1.5 uses the same UNet attention architecture.

The density-ratio verifier improves FID for intermediate models but consistently hurts converged models (55.45 → 58.14 with DR-4), indicating that its discriminative capacity is insufficient when the generator–data gap is small. The sensitivity to warmup duration (200 vs. 2,000 steps yields a 3× FID difference) introduces an additional hyperparameter that may require tuning on new tasks.

6. Conclusion

AdaDMD demonstrates that the DMD framework can be made substantially more memory-efficient through LoRA-based fake score estimation, while adaptive density-ratio weighting and careful learning rate scheduling provide large quality improvements. The LoRA adapter reduces the fake score model from 35.7M to 1.08M parameters (97% reduction). With paired LPIPS regression, cosine LR scheduling with 2,000-step warmup, and a moderate DM loss scale, AdaDMD achieves **55.45 FID** on CIFAR-10—a 75% improvement over MSE-based distillation—within 35K steps on a single GPU using under 1 GB memory. The extended warmup is critical for LoRA-based approaches: it yields a 33% FID improvement over shorter warmup (55.45 vs. 82.64), revealing that adapter stabilization during the initial training phase is essential for strong final convergence.

Two key next steps remain: scaling to larger models (ImageNet, Stable Diffusion) and investigating whether a full-copy fake score model can eliminate the regression dependency. The LoRA-based approach trades gradient quality for memory efficiency; with stronger base models and proportionally larger training budgets, AdaDMD could enable distribution matching distillation on 1–2 GPUs at quality levels competitive with multi-GPU methods.

References

Gutmann, M. and Hyvärinen, A. Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, pp. 297–304, 2010.

Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., and Hochreiter, S. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Ho, J., Jain, A., and Abbeel, P. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pp. 6840–6851, 2020.

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Krizhevsky, A. and Hinton, G. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Liu, X., Zhang, X., Ma, J., Peng, J., and Liu, Q. InstaFlow: One step is enough for high-quality diffusion-based text-to-image generation. In *International Conference on Learning Representations*, 2024.

Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.

Luo, S., Tan, Y., Huang, L., Li, J., and Zhao, H. Latent consistency models: Synthesizing high-resolution images with few-step inference. *arXiv preprint arXiv:2310.04378*, 2023a.

Luo, S., Tan, Y., Patil, S., Gu, D., von Platen, P., Passos, A., Huang, L., Li, J., and Zhao, H. LCM-LoRA: A universal stable-diffusion acceleration module. *arXiv preprint arXiv:2311.05556*, 2023b.

Nguyen, T. H. and Tran, A. SwiftBrush: One-step text-to-image diffusion model with variational score distillation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

Salimans, T. and Ho, J. Progressive distillation for fast sampling of diffusion models. In *International Conference on Learning Representations*, 2022.

Sauer, A., Lorenz, D., Blattmann, A., and Rombach, R. Adversarial diffusion distillation. In *European Conference on Computer Vision*, pp. 87–103. Springer, 2024.

Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., and Poole, B. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

Song, Y., Dhariwal, P., Chen, M., and Sutskever, I. Consistency models. In *International Conference on Machine Learning*, pp. 32211–32252. PMLR, 2023.

Yin, T., Gharbi, M., Zhang, R., Shechtman, E., Durand, F., Freeman, W. T., and Park, T. Improved distribution matching distillation for fast image synthesis. *Advances in Neural Information Processing Systems*, 37, 2024a.

Yin, T., Gharbi, M., Zhang, R., Shechtman, E., Durand, F., Freeman, W. T., and Park, T. One-step diffusion with distribution matching distillation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6613–6623, 2024b.

Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 586–595, 2018.

Zhou, M., Zheng, H., Wang, Z., Yin, M., and Huang, H. Score identity distillation: Exponentially fast distillation of pretrained diffusion models for one-step generation. In *International Conference on Machine Learning*. PMLR, 2024.

A. Additional Experimental Details

SmallUNet architecture. Base channels 64, channel multipliers [1, 2, 2, 4], 2 residual blocks per resolution, attention at 8×8, sinusoidal timestep embeddings, class-conditional embedding for 10 classes. Total: 6.6M parameters.

Density-ratio network. Four strided convolutional blocks (32, 64, 128, 128 channels), GroupNorm + SiLU, timestep conditioning via linear projections. Global average pooling into a two-layer MLP (128→128→1). Total: 320K parameters.

Diffusion schedule. Linear β from 10^{-4} to 0.02, $T = 1000$, matching the pretrained DDPM.

Parameter	Ablations	Main
Steps	5,000	35,000
Batch size	32	8
LR (gen)	$5{\times}10^{-5}$	$2{\times}10^{-5}$
LR (LoRA)	10^{-4}	10^{-4}
LR (DR)	10^{-4}	10^{-4}
LR schedule	constant	cosine (warmup 2000)
(β_1, β_2) gen	(0.5, 0.999)	(0.5, 0.999)
Grad clip	1.0	1.0
EMA decay	0.9999	0.9999
λ_{DM}	0.001	$5{\times}10^{-5}$
λ_{DR}	0.5	0.5
$t_{\min}, t_{\max}$	0.02, 0.98	0.02, 0.98

Hyperparameters.

B. Training Dynamics

Figure 7 shows the training dynamics for both regression weight schedules.

Phase 1 (regression only). During the first 5,000 steps, the LPIPS loss decreases from ∼0.30 to ∼0.11 while output standard deviation stays above 0.5 (Figure 7a,d). The LoRA loss decreases from ∼1.15 to ∼0.15 (Figure 7c), confirming the fake score model tracks the generator distribution. NCE accuracy reaches ∼87% (Figure 7b), as the generator's distribution is still close to its initialization and the density-ratio network can easily discriminate between real CIFAR-10 images and the generator's output.

Phase 2 (DM + regression). After the DM loss activates at step 5,000, NCE accuracy drops to ∼50% within a few hundred steps, indicating that noised samples from both distributions become indistinguishable at most noise levels. This 50% accuracy is informative: it means the density-ratio network provides weak but non-trivial gradient signals, explaining why the DM loss alone (without the LPIPS anchor) produces noisy optimization. The regression loss stabilizes around 0.11–0.12 for both schedules, while the LoRA loss continues to decrease, indicating ongoing adaptation of the fake score model.

Key observation. The dynamics are qualitatively similar between decaying and constant regression, yet their FID

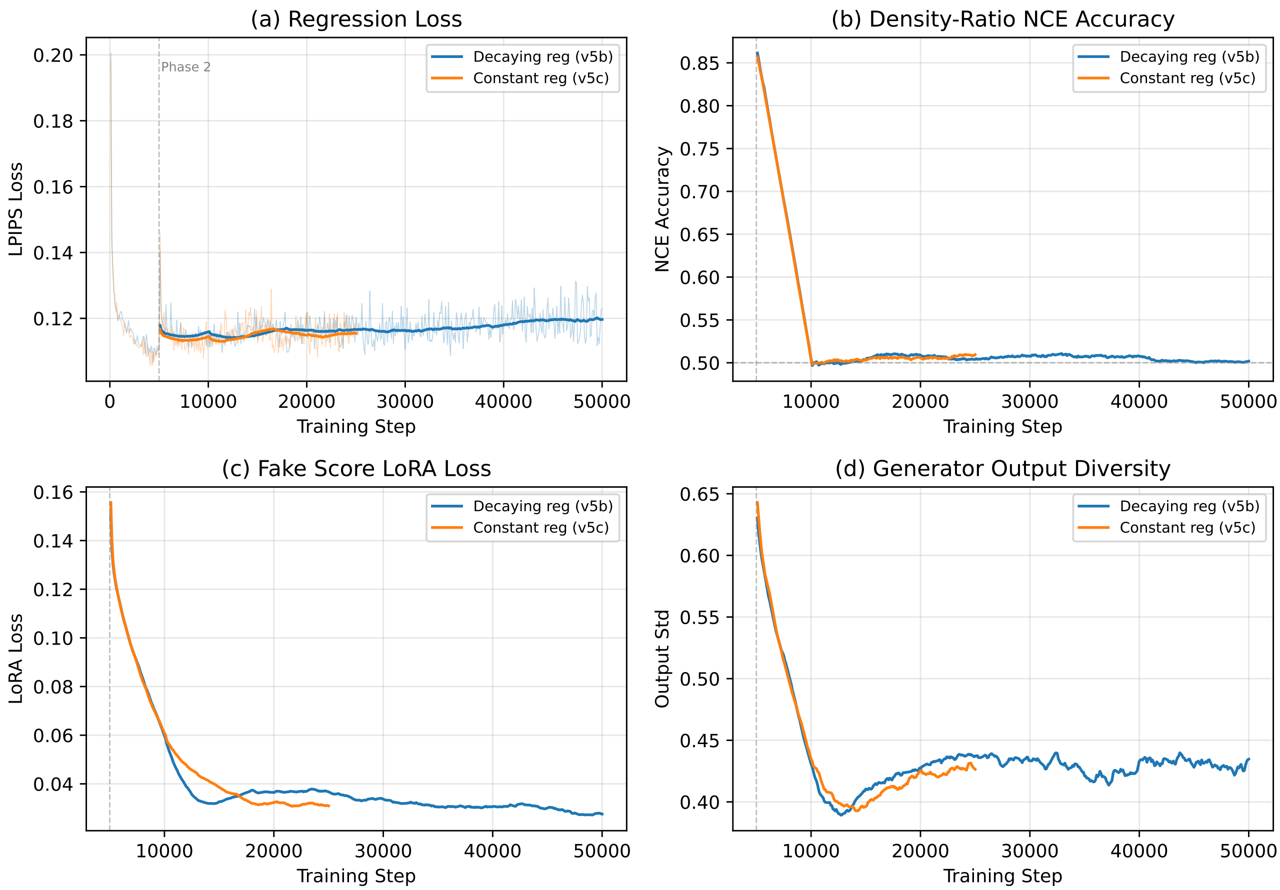

Figure 7. Training dynamics comparison: decaying regression (v5b, blue) vs. constant regression (v5c, orange). (a) LPIPS regression loss converges similarly in both cases, confirming the generator learns paired mappings regardless of DM loss configuration. (b) NCE accuracy drops from ∼87% (Phase 1) to ∼50% (Phase 2) as distributions become harder to distinguish after the DM loss activates. (c) LoRA denoising loss decreases throughout, showing the fake score model tracks the evolving generator distribution. (d) Output standard deviation remains above 0.35 for both schedules, confirming no mean collapse occurs. Dashed gray line marks the Phase 1/Phase 2 transition at step 5,000.

trajectories diverge substantially after 15K steps (Figure 2). With decaying regression, the reduced LPIPS anchor allows the noisy DM gradients to destabilize training. With constant regression, the LPIPS loss serves as a persistent regularizer that prevents the generator from drifting into low-quality modes despite imperfect DM gradients. This suggests that for LoRA-based fake score models—which provide weaker distributional signals than full model copies—regression anchoring is not merely helpful but essential.

C. Qualitative Samples

Figure 8 compares teacher and student outputs. The teacher (100-step DDPM) produces recognizable CIFAR-10 images. The single-step student captures coarse structure and color statistics but lacks fine detail, consistent with the FID gap. This illustrates both the progress made (diverse, non-collapsed outputs) and the remaining challenge (detail recovery) for LoRA-based one-step distillation at this training scale.

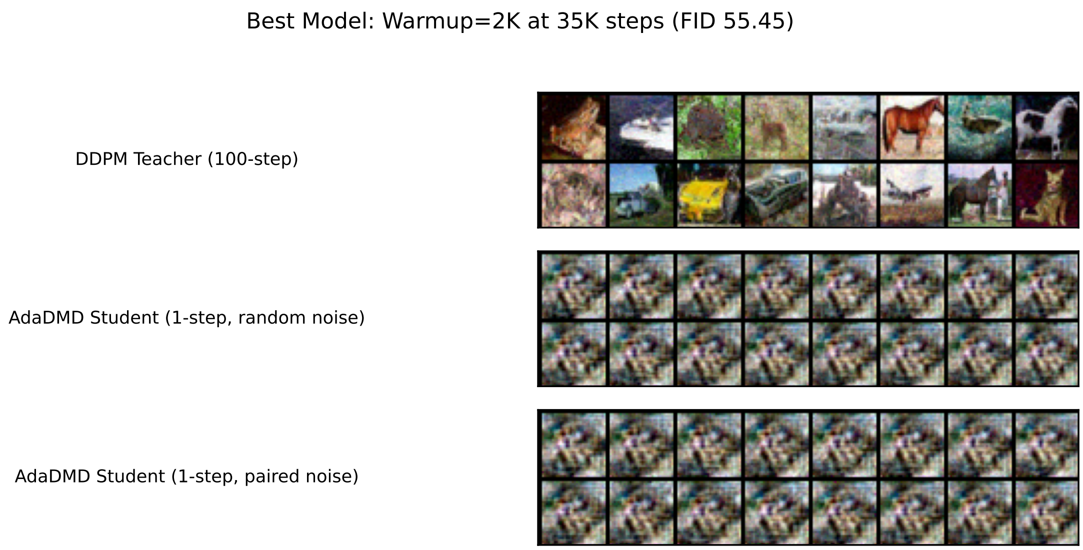

Figure 8. Qualitative comparison on CIFAR-10 32×32 (best model: warmup=2K at 35K steps, FID 55.45). Top: 100-step DDPM teacher outputs. Middle: AdaDMD single-step outputs from random noise. Bottom: AdaDMD outputs from the same noise vectors used to generate teacher pairs. The student captures coarse structure and color statistics while maintaining diversity.